WORKED SOLUTION

STEP III 1996, Question 13

Probability and Statistics • Chebyshev's Inequality

STEP III 1996, Question 13

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/96-s3-q13

The key idea

The variance is a sum of non-negative terms $p_i(x_i - \mu)^2$, one for each value. Split that sum into the values that are far from the mean (at least λ away) and those that are near. For an *upper* bound on the far probability, throw the near terms away and bound each far term below. For a *lower* bound, do the opposite: bound every term above, the near ones by λ^2 and the far ones by k^2.

Solution

Definitions and $aX + b$

Write $p_i = \mathrm{P}(X = x_i)$, so that $p_i \geq 0$ and $\sum_{i=1}^{n} p_i = 1$. Then

$$\mathrm{E}(X) = \sum_{i=1}^{n} x_i p_i, \qquad \mathrm{var}(X) = \mathrm{E}((X-\mu)^2) = \sum_{i=1}^{n} (x_i - \mu)^2 p_i,$$

where $\mu = \mathrm{E}(X)$. (Equivalently $\mathrm{var}(X) = \mathrm{E}(X^2) - \mu^2$, but the first form is the one we shall use.)

The random variable $aX + b$ takes the value $ax_i + b$ with probability p_i. (If $a = 0$ these values coincide, but adding the probabilities of equal values does not change the sums below.) So

$$\mathrm{E}(aX + b) = \sum_i (ax_i + b)p_i = a\sum_i x_i p_i + b\sum_i p_i = a\mathrm{E}(X) + b,$$

using $\sum_i p_i = 1$. Hence $aX + b - \mathrm{E}(aX + b) = a(X - \mu)$, and

$$\mathrm{var}(aX + b) = \sum_i \left(a(x_i - \mu)\right)^2 p_i = a^2 \sum_i (x_i - \mu)^2 p_i = a^2\,\mathrm{var}(X).$$

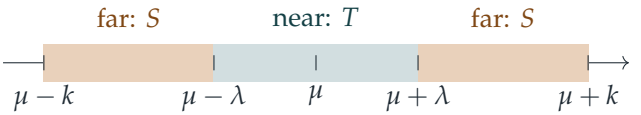
Adding b shifts every value and the mean together, so it has no effect on the spread.

The upper bound

Let $\mu = \mathrm{E}(X)$, let S be the set of indices i with $|x_i - \mu| \geq \lambda$ (the "far" values), and let T be the rest, where $|x_i - \mu| < \lambda$. Then

$$\mathrm{P}(|X - \mu| \geq \lambda) = \sum_{i \in S} p_i.$$

Call this probability P.



Every term of the variance is non-negative, so dropping the terms in T can only make the sum smaller; and for $i \in S$ we have $(x_i - \mu)^2 \geq \lambda^2$. Therefore

$$\begin{aligned}\mathrm{var}(X) &= \sum_{i \in S} (x_i - \mu)^2 p_i + \sum_{i \in T} (x_i - \mu)^2 p_i \\ &\geq \sum_{i \in S} (x_i - \mu)^2 p_i \\ &\geq \lambda^2 \sum_{i \in S} p_i = \lambda^2 P.\end{aligned}$$

Since $\lambda > 0$ we may divide by λ^2, giving

$$\mathrm{P}(|X - \mu| \geq \lambda) \leq \frac{\mathrm{var}(X)}{\lambda^2},$$

as required. (If S is empty both sums over S are zero and the inequality is trivially true.)

The lower bound

Now suppose also that $|x_i - \mu| \leq k$ for every i. This time we bound every term of the variance from *above*:

- for $i \in S$, $(x_i - \mu)^2 \leq k^2$ by the new hypothesis;
- for $i \in T$, $(x_i - \mu)^2 < \lambda^2$ by the definition of T.

Since $\sum_{i \in T} p_i = 1 - P$, this gives

$$\mathrm{var}(X) \leq k^2 \sum_{i \in S} p_i + \lambda^2 \sum_{i \in T} p_i = k^2 P + \lambda^2 (1 - P) = \lambda^2 + (k^2 - \lambda^2) P.$$

Rearranging,

$$(k^2 - \lambda^2)\, P \geq \mathrm{var}(X) - \lambda^2.$$

Ask yourself Can we divide by $k^2 - \lambda^2$?

We are told $k \geq \lambda > 0$, so $k^2 - \lambda^2 \geq 0$. If $k > \lambda$ it is strictly positive, and dividing keeps the direction of the inequality:

$$\mathrm{P}\big(|X - \mu| \geq \lambda\big) \geq \frac{\mathrm{var}(X) - \lambda^2}{k^2 - \lambda^2},$$

which is the required result. In the boundary case $k = \lambda$ the right-hand side has a zero denominator and the statement has no meaning; the inequality just before it still holds and says $\mathrm{var}(X) \leq \lambda^2$, which is clear since every $(x_i - \mu)^2 \leq k^2 = \lambda^2$.

When $\mathrm{var}(X) \leq \lambda^2$ the lower bound is zero or negative and so says nothing new; it has content only when the variance is large compared with λ^2, forcing some probability out into the tails.

Checks

- Take $X = \pm 1$ with probability 0.4 each and ± 4 with probability 0.1 each, so $\mu = 0$ and $\mathrm{var}(X) = 0.8 \times 1 + 0.2 \times 16 = 4$. With $\lambda = 1$, $k = 4$: $P = 1$, the upper bound is 4 and the lower bound is $\frac{3}{15} = \frac{1}{5}$, and indeed $\frac{1}{5} \leq 1 \leq 4$.
- Take $X = 0$ or $\pm c$, with $\mathrm{P}(X = \pm c) = q/2$ each. Then $\mathrm{var}(X) = qc^2$ and with $\lambda = c$ the upper bound is exactly $q = P$: Chebyshev's inequality cannot be improved in general.
- We tested both bounds, and the formulae for $\mathrm{E}(aX + b)$ and $\mathrm{var}(aX + b)$, in exact rational arithmetic on 20 000 random distributions with up to seven values; no case failed.

Common slips

- Writing $\mathrm{var}(aX + b) = a\,\mathrm{var}(X)$ or $a^2\mathrm{var}(X) + b$: the variance is quadratic in deviations, and a shift changes no deviation.
- In the lower bound, bounding the near terms by 0 instead of λ^2. That gives $\mathrm{var}(X) \leq k^2 P$, which is true but not what is asked.
- Dividing by $k^2 - \lambda^2$ without remarking that it is positive (and that $k = \lambda$ must be excluded).

Beyond STEP

The first inequality is Chebyshev's inequality, and the trick of discarding terms is the same one behind Markov's inequality $\mathrm{P}(Y \geq c) \leq \mathrm{E}(Y)/c$ for $Y \geq 0$ (apply it to $Y = (X - \mu)^2$). In a first-year probability course Chebyshev's inequality gives a two-line proof of the weak law of large numbers: the average of n independent copies of X has variance $\mathrm{var}(X)/n$, so it is unlikely to stray far from μ. The second inequality is a "reverse" version that needs boundedness, in the same spirit as the Paley–Zygmund inequality. Grimmett and Welsh, *Probability: An Introduction*, covers all of this.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.